



IoT-Based DATA LOGGER Using ThingSpeak And WeMos

SOMNATH BERA

This project is a data-logging system that displays temperature and humidity on *ThingSpeak.com* using WeMos D1 R2 board. You can add more data to the project—up to eight data per channel.

WeMos board

WeMos D1 R2 board looks and works almost like Arduino Uno, as shown in Fig. 1. It is an ESP8266 development board programmable via Arduino IDE. The board is based on a 32-bit processor, having 160MHz clock speed and 4Mb flash memory. It is 17 times faster than Arduino Uno. WeMos works on 3.3V, while Arduino Uno works on 3.3V to 5V.



Fig. 1: WeMos D1 R2 board

Channel Settings

Percentage complete 65%

Channel ID 279012

Name Bera Data channel

Description Bera ESP8266 channel

Field 1 Humidity ☒

Field 2 Temperature ☒

Field 3 Humidity ☒

Field 4 ☐

Fig. 2: Channel ID and description

In this project we set WeMos to operate at lowest possible power mode when inactive. In active mode, it takes 70mA to 80mA current from a 3.7V lithium-ion cell. It draws about 14mA during sleep mode for 15 to 20 seconds between consecutive uploads. It draws 8mA for three minutes during pause periods.

ThingSpeak IoT platform

ThingSpeak.com is an open source Internet of Things (IoT) application and API to store and retrieve data from things using HTTP protocol over the Internet via LAN. ThingSpeak enables creation of sensor-logging applications, location-tracking applications and a social network of things with status updates. This means, if you can send data from your sensors to ThingSpeak at regular intervals, it will create, store and display data

in a trend automatically. It also has Maths modelling inbuilt in the form of MATLAB documentation, which you can use for free.

Getting started with ThingSpeak

Create account and set up channel.

The first thing you need to do is to open an account and a channel on *ThingSpeak.com*. For that, you need an email account. *ThingSpeak.com* will send a confirmation email to your email address. Click on the link in the email to validate your account, and the account will be opened. Now, you can create as many channels as you want.

Once you create a channel, you will get three ThingSpeak identities: channel ID, write API key and read API key. Note these down as you will require these later. Meanwhile, assign a name to each data field in

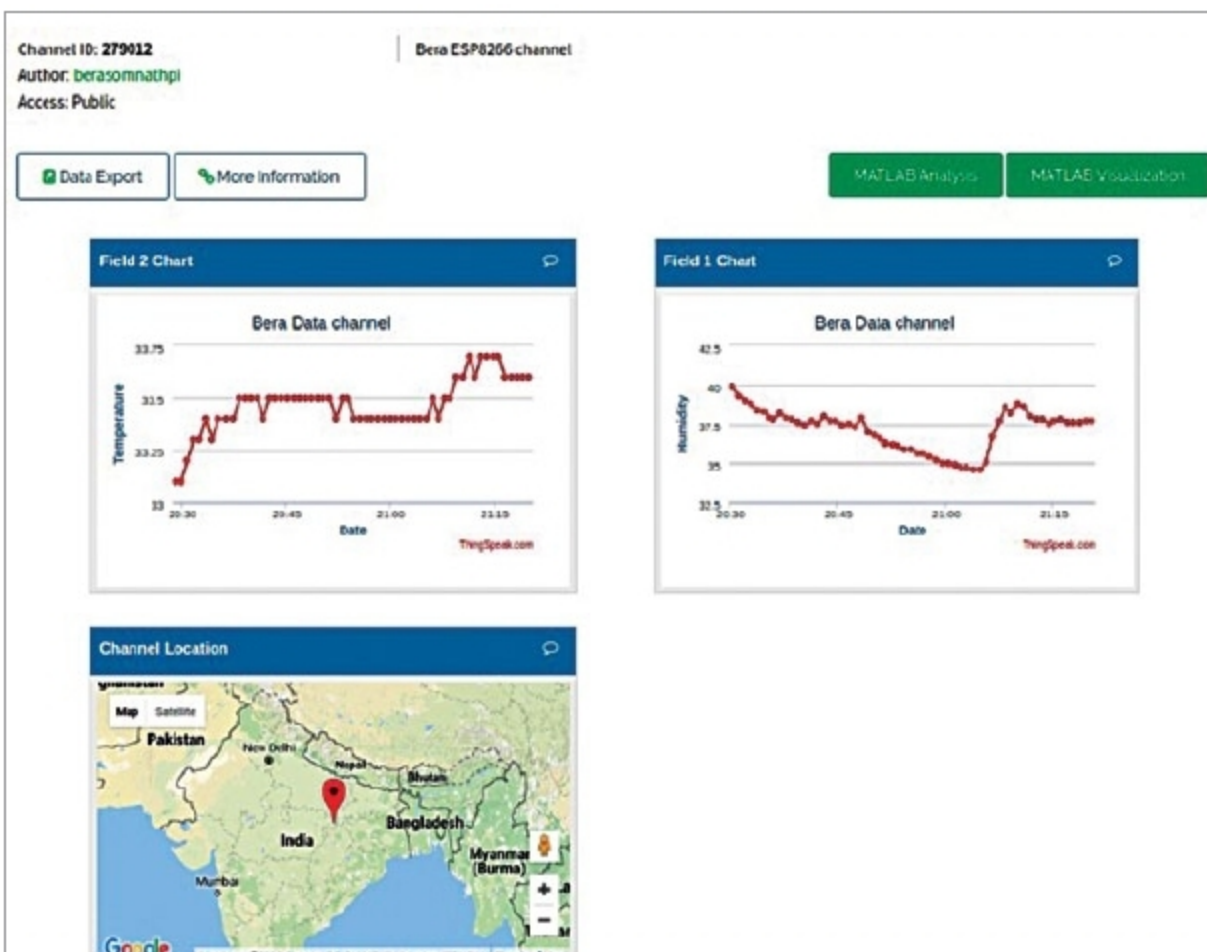


Fig. 3: Data on ThingSpeak.com